

Team 201 - DQNDDQN

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0.1 Autonomous Drone Battery Management for Urban Surveillance using DQN and DDQN - 7 Marks

0.1.1 Import Statements

```
[1]: import numpy as np
import tensorflow as tf
from collections import deque
import random
import matplotlib.pyplot as plt
import time
import os
os.environ['TF_CPP_MIN_LOG_LEVEL'] = '2' # Reduce TensorFlow logging
import tensorflow as tf
tf.config.experimental.enable_tensor_float_32_execution(False)
```

```
[8]: # --- 1. Hyperparameters and Constants ---

# Environment Parameters
GRID_SIZE = (10, 10) # X, Y
BATTERY_CAPACITY = 100.0 # Percentage
INITIAL_BATTERY = 100.0

BASE_MOVE_COST = 0.5 # % battery per step
BASE_HOVER_COST = 0.2 # % battery per step
RECHARGE_RATE = 5.0 # % battery per step when recharging

BATTERY_CRASH_PENALTY = -100.0
TIME_PENALTY = -0.1 # Small penalty for each time step
RECHARGE_BONUS = 1.0 # Small bonus for successfully recharging
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# POI Parameters
POI_SPAWN_CHANCE = 0.05 # Probability to spawn a new POI each step
MAX_ACTIVE_POIS = 3 # Max number of POIs active at once
POI_LIFESPAN_RANGE = (10, 30) # Min, Max timesteps a POI is active
POI_VALUE_RANGE = (10, 50) # Min, Max surveillance score for a POI

# Atmospheric Disturbance Parameters
DISTURBANCE_FACTOR = 0.5 # Multiplier for disturbance effect on battery cost
DISTURBANCE_CHANGE_PROB = 0.1 # Probability disturbance changes randomly each
↳step
DISTURBANCE_MAGNITUDE_CHANGE = 0.1 # Max change when disturbance updates

# Agent Parameters
STATE_SIZE = 7 # [drone_x, drone_y, battery_level, disturbance,
↳nearest_poi_dist, nearest_poi_value, nearest_poi_lifespan]
ACTION_SIZE = 6 # North, South, East, West, Hover, Recharge

LEARNING_RATE = 0.001
DISCOUNT_FACTOR = 0.95 # Gamma

REPLAY_BUFFER_SIZE = 5000
MIN_REPLAY_SIZE = 500 # Start learning after this many experiences
BATCH_SIZE = 32

EXPLORATION_MAX = 1.0
EXPLORATION_MIN = 0.01
EXPLORATION_DECAY = 0.995 # Decay epsilon over time

TARGET_UPDATE_FREQUENCY = 50 # How often to update the target network

# Training Parameters
EPISODES = 50 # Number of training episodes (24-hour cycles)
MAX_TIMESTEPS_PER_EPISODE = 200 # Represents a 24-hour cycle (or portion)

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[9]: # --- 2. Replay Buffer Class ---
class ReplayBuffer:
    """Experience replay buffer for storing and sampling experiences"""
    def __init__(self, capacity):
        self.buffer = deque(maxlen=capacity)

    def push(self, state, action, reward, next_state, done):
        """Store an experience tuple"""
        self.buffer.append((state, action, reward, next_state, done))

    def sample(self, batch_size):
        """Sample a batch of experiences"""

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        batch = random.sample(self.buffer, batch_size)
        states, actions, rewards, next_states, dones = zip(*batch)
        return (np.array(states), np.array(actions), np.array(rewards),
                np.array(next_states), np.array(dones))

    def __len__(self):
        return len(self.buffer)

```

0.1.2 — 3. Custom Environment: DroneSurveillanceEnv — - 2 Marks

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[10]: # Develop the DroneSurveillanceEnv including the __init__, step,
# and reset methods. Accurately model battery dynamics (depletion, recharge,
↳ capacity limits).
# Implement dynamic POI spawning, value collection, and lifespan decay.
# Model random atmospheric disturbances and their effect on energy consumption.

class DroneSurveillanceEnv:
    """Custom environment for drone surveillance with battery management"""

    def __init__(self, grid_size, battery_capacity, initial_battery):
        # Grid and basic parameters
        self.grid_size = grid_size
        self.battery_capacity = battery_capacity
        self.initial_battery = initial_battery

        # Drone State
        self.drone_pos = [0, 0] # [x, y]
        self.battery_level = initial_battery

        # Environmental Dynamics
        self.atmospheric_disturbance = 0.0 # 0.0 to 1.0
        self.active_pois = [] # List of active POIs: [(x, y, value, lifespan),
↳ ...]

        self.timestep = 0

        # Charging Stations (fixed locations) - corners of the grid
        self.charging_stations = [(0, 0), (0, grid_size[1]-1),
                                  (grid_size[0]-1, 0), (grid_size[0]-1,
↳ grid_size[1]-1)]

        # Action mapping: 0:N, 1:S, 2:E, 3:W, 4:Hover, 5:Recharge
        self.action_map = {
            0: (0, 1), # North
            1: (0, -1), # South
            2: (1, 0), # East
            3: (-1, 0), # West
            4: (0, 0), # Hover

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        5: (0, 0)      # Recharge (special case)
    }

    def reset(self):
        """Reset the environment to initial state"""
        # Reset drone position to random charging station
        self.drone_pos = list(random.choice(self.charging_stations))
        self.battery_level = self.initial_battery
        self.atmospheric_disturbance = random.uniform(0.0, 0.5)
        self.active_pois = []
        self.timestep = 0
        return self._get_obs()

    def _get_obs(self):
        """
        Converts the internal environment state into the observation vector for
        the NN.
        Returns: [drone_x, drone_y, battery_level, disturbance,
        nearest_poi_dist, nearest_poi_value, nearest_poi_lifespan]
        """
        # Basic state information
        obs = [
            self.drone_pos[0] / (self.grid_size[0] - 1), # Normalized drone x
            self.drone_pos[1] / (self.grid_size[1] - 1), # Normalized drone y
            self.battery_level / self.battery_capacity,   # Normalized battery
            self.atmospheric_disturbance,                 # Disturbance level
        ]

        # --- POI Information ---
        if self.active_pois:
            # Find nearest POI
            min_dist = float('inf')
            nearest_poi = None
            for poi in self.active_pois:
                dist = np.sqrt((self.drone_pos[0] - poi[0])**2 + (self.
                drone_pos[1] - poi[1])**2)
                if dist < min_dist:
                    min_dist = dist
                    nearest_poi = poi

            # Normalize POI information
            max_dist = np.sqrt(self.grid_size[0]**2 + self.grid_size[1]**2)
            obs.extend([
                min_dist / max_dist, # Normalized distance to nearest POI
                nearest_poi[2] / POI_VALUE_RANGE[1], # Normalized POI value
                nearest_poi[3] / POI_LIFESPAN_RANGE[1] # Normalized POI
                lifespan

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    ])
else:
    # No active POIs
    obs.extend([1.0, 0.0, 0.0]) # Max distance, no value, no lifespan

    return np.array(obs, dtype=np.float32)

def _spawn_poi(self):
    """Randomly spawn a new POI if conditions are met"""
    if len(self.active_pois) >= MAX_ACTIVE_POIS:
        return

    if random.random() < POI_SPAWN_CHANCE:
        # Find empty cell
        attempts = 0
        while attempts < 50: # Prevent infinite loop
            x = random.randint(0, self.grid_size[0] - 1)
            y = random.randint(0, self.grid_size[1] - 1)

            # Check if cell is occupied by drone, charging station, or
            ↪existing POI
            occupied = False
            if [x, y] == self.drone_pos:
                occupied = True
            if (x, y) in self.charging_stations:
                occupied = True
            for poi in self.active_pois:
                if poi[0] == x and poi[1] == y:
                    occupied = True
                    break

            if not occupied:
                value = random.randint(*POI_VALUE_RANGE)
                lifespan = random.randint(*POI_LIFESPAN_RANGE)
                self.active_pois.append((x, y, value, lifespan))
                break

        attempts += 1

def _update_pois(self, drone_at_poi_pos):
    """Update POI lifespans and handle collection"""
    reward = 0
    new_pois = []

    for poi in self.active_pois:
        x, y, value, lifespan = poi

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        # Check if drone collected this POI
        if drone_at_poi_pos and self.drone_pos[0] == x and self.
↪drone_pos[1] == y:
            reward += value # Collect POI reward
            continue # POI disappears after collection

        # Decrease lifespan
        new_lifespan = lifespan - 1
        if new_lifespan > 0:
            new_pois.append((x, y, value, new_lifespan))

    self.active_pois = new_pois
    return reward

def _update_atmospheric_disturbance(self):
    """Update atmospheric disturbance with random walk"""
    if random.random() < DISTURBANCE_CHANGE_PROB:
        change = random.uniform(-DISTURBANCE_MAGNITUDE_CHANGE,
↪DISTURBANCE_MAGNITUDE_CHANGE)
        self.atmospheric_disturbance = np.clip(self.atmospheric_disturbance,
↪+ change, 0.0, 1.0)

def step(self, action):
    """Execute one step in the environment"""
    reward = 0
    done = False
    info = {}

    # --- Update Atmospheric Disturbance ---
    self._update_atmospheric_disturbance()

    # --- Battery Consumption / Recharge ---
    battery_cost = 0

    if action == 5: # Recharge action
        if tuple(self.drone_pos) in self.charging_stations:
            # Successfully recharging
            self.battery_level = min(self.battery_capacity,
↪self.battery_level + RECHARGE_RATE)
            reward += RECHARGE_BONUS
        else:
            # Can't recharge, treat as hover
            battery_cost = BASE_HOVER_COST + (self.atmospheric_disturbance,
↪* DISTURBANCE_FACTOR)
            elif action == 4: # Hover
                battery_cost = BASE_HOVER_COST + (self.atmospheric_disturbance *
↪DISTURBANCE_FACTOR)

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        else: # Move actions
            battery_cost = BASE_MOVE_COST + (self.atmospheric_disturbance *
↳DISTURBANCE_FACTOR)

            # Apply movement
            dx, dy = self.action_map[action]
            new_x = np.clip(self.drone_pos[0] + dx, 0, self.grid_size[0] - 1)
            new_y = np.clip(self.drone_pos[1] + dy, 0, self.grid_size[1] - 1)
            self.drone_pos = [new_x, new_y]

            # Apply battery cost
            self.battery_level -= battery_cost

            # --- Check for Crash ---
            if self.battery_level <= 0 and tuple(self.drone_pos) not in self.
↳charging_stations:
                reward += BATTERY_CRASH_PENALTY
                done = True
                info['crash'] = True

            # --- Update POIs and Collect Rewards ---
            poi_reward = self._update_pois(not done) # Don't collect if crashed
            reward += poi_reward

            # --- Spawn new POIs randomly ---
            self._spawn_poi()

            # --- Add time penalty ---
            reward += TIME_PENALTY

            # --- Check episode termination ---
            self.timestep += 1
            if self.timestep >= MAX_TIMESTEPS_PER_EPISODE:
                done = True

            return self._get_obs(), reward, done, info

def render(self):
    """Render the current state of the environment"""
    grid = np.zeros(self.grid_size, dtype=str)
    grid[:] = '.'

    # Mark Charging Stations
    for cs in self.charging_stations:
        grid[cs[0], cs[1]] = 'C'

    # Mark Active POIs

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    for poi in self.active_pois:
        grid[poi[0], poi[1]] = 'P'

    # Mark Drone
    grid[self.drone_pos[0], self.drone_pos[1]] = 'D'

    # Print grid
    print(f"Battery: {self.battery_level:.1f}%, Disturbance: {self.
↪atmospheric_disturbance:.2f}%")
    print(" ", end=" ")
    for j in range(self.grid_size[1]):
        print(f"{j}", end=" ")
    print()

    for i in range(self.grid_size[0]):
        print(f"{i} ", end=" ")
        for j in range(self.grid_size[1]):
            print(grid[i, j], end=" ")
        print()
    print()

```

0.1.3 — 4. DQNAgent Class — 1 Mark

```

[11]: class DQNAgent:
    """Deep Q-Network agent with optional Double DQN capability"""

    def __init__(self, state_size, action_size, use_ddqn=False):
        self.state_size = state_size
        self.action_size = action_size
        self.use_ddqn = use_ddqn

        # Experience replay
        self.memory = ReplayBuffer(REPLAY_BUFFER_SIZE)

        # Exploration parameters
        self.epsilon = EXPLORATION_MAX
        self.epsilon_min = EXPLORATION_MIN
        self.epsilon_decay = EXPLORATION_DECAY

        # Neural networks
        self.q_network = self._build_model()
        self.target_q_network = self._build_model()

        # Initialize target network
        self.update_target_network()

        # Training step counter

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self.train_step = 0

def _build_model(self):
    """
    Builds the neural network for the Q-function.
    Input: State vector (STATE_SIZE)
    Output: Q-values for each action (ACTION_SIZE)
    """
    model = tf.keras.Sequential([
        tf.keras.layers.Dense(64, activation='relu', input_shape=(self.
↪state_size,)), # Reduce from 128
        tf.keras.layers.Dense(64, activation='relu'), # Reduce from 128
        tf.keras.layers.Dense(32, activation='relu'), # Reduce from 64
        tf.keras.layers.Dense(self.action_size, activation='linear')
    ])

    model.compile(optimizer=tf.keras.optimizers.
↪Adam(learning_rate=LEARNING_RATE),
                loss='mse')
    return model

def update_target_network(self):
    """Copy weights from main network to target network"""
    self.target_q_network.set_weights(self.q_network.get_weights())

def choose_action(self, state):
    """Choose action using epsilon-greedy policy"""
    if np.random.random() <= self.epsilon:
        return random.randrange(self.action_size)

    q_values = self.q_network.predict(state.reshape(1, -1), verbose=0)
    return np.argmax(q_values[0])

def remember(self, state, action, reward, next_state, done):
    """Store experience in replay buffer"""
    self.memory.push(state, action, reward, next_state, done)

def learn(self):
    """Train the network on a batch of experiences"""
    if len(self.memory) < MIN_REPLAY_SIZE:
        return

    if len(self.memory) < BATCH_SIZE:
        return

    # Sample batch from memory

```

```

        states, actions, rewards, next_states, dones = self.memory.
        ↪sample(BATCH_SIZE)

        # Current Q values
        current_q_values = self.q_network.predict(states, verbose=0)

        # Next Q values from target network
        if self.use_ddqn:
            # Double DQN: Use main network to select actions, target network to
            ↪evaluate
            next_q_values_main = self.q_network.predict(next_states, verbose=0)
            next_q_values_target = self.target_q_network.predict(next_states,
            ↪verbose=0)

            # Select actions using main network
            next_actions = np.argmax(next_q_values_main, axis=1)

            # Evaluate selected actions using target network
            next_q_values = next_q_values_target[np.arange(BATCH_SIZE),
            ↪next_actions]
        else:
            # Standard DQN: Use target network for both selection and evaluation
            next_q_values_target = self.target_q_network.predict(next_states,
            ↪verbose=0)
            next_q_values = np.max(next_q_values_target, axis=1)

        # Calculate target Q values
        target_q_values = rewards + (DISCOUNT_FACTOR * next_q_values * (1 -
        ↪dones))

        # Update current Q values
        for i in range(BATCH_SIZE):
            current_q_values[i][actions[i]] = target_q_values[i]

        # Train the network
        self.q_network.fit(states, current_q_values, epochs=1, verbose=0)

        # Decay epsilon
        if self.epsilon > self.epsilon_min:
            self.epsilon *= self.epsilon_decay

        # Update target network periodically
        self.train_step += 1
        if self.train_step % TARGET_UPDATE_FREQUENCY == 0:
            self.update_target_network()

```

0.1.4 — 5. Main Training Loop — 1 Mark

```
[12]: def train_agent(env, agent, num_episodes, max_timesteps_per_episode,
    ↪render=False):
    """Train the DQN agent"""
    scores = []
    epsilon_history = []

    for episode in range(num_episodes):
        state = env.reset()
        total_reward = 0
        steps_taken = 0

        for timestep in range(max_timesteps_per_episode):
            # Choose and execute action
            action = agent.choose_action(state)
            next_state, reward, done, info = env.step(action)

            # Store experience
            agent.remember(state, action, reward, next_state, done)

            # Learn from experience (only if we have enough experiences)
            if len(agent.memory) > MIN_REPLAY_SIZE:
                agent.learn()

            total_reward += reward
            state = next_state
            steps_taken += 1

            if done:
                break

        scores.append(total_reward)
        epsilon_history.append(agent.epsilon)

        # Print progress more frequently for faster feedback
        if episode % 10 == 0: # Change from 50 to 10
            avg_score = np.mean(scores[-10:]) if len(scores) >= 10 else np.
    ↪mean(scores)
            print(f"Episode {episode}, Average Score: {avg_score:.2f}, Epsilon:
    ↪{agent.epsilon:.3f}, Steps: {steps_taken}")

    return scores, epsilon_history

def evaluate_agent(env, agent, num_episodes=10):
    """Evaluate the trained agent"""
```

```

scores = []

# Save current epsilon and set to 0 for evaluation
original_epsilon = agent.epsilon
agent.epsilon = 0.0

for episode in range(num_episodes):
    state = env.reset()
    total_reward = 0

    for timestep in range(MAX_TIMESTEPS_PER_EPISODE):
        action = agent.choose_action(state)
        next_state, reward, done, info = env.step(action)
        total_reward += reward
        state = next_state

        if done:
            break

    scores.append(total_reward)

# Restore epsilon
agent.epsilon = original_epsilon

return scores

```

0.1.5 — Main Execution Block —

```

[14]: if __name__ == "__main__":
    # Ensure TensorFlow uses GPU if available
    print("TensorFlow version:", tf.__version__)
    print("GPU Available:", tf.config.list_physical_devices('GPU'))

    # Initialize environment
    env = DroneSurveillanceEnv(GRID_SIZE, BATTERY_CAPACITY, INITIAL_BATTERY)

    # --- Train DQN Agent ---
    print("Training DQN Agent...")
    dqn_agent = DQNAgent(STATE_SIZE, ACTION_SIZE, use_ddqn=False)
    dqn_scores, dqn_epsilon = train_agent(env, dqn_agent, EPISODES,
    ↪MAX_TIMESTEPS_PER_EPISODE)

    # --- Train Double DQN Agent --- 1 Mark
    print("\nTraining Double DQN Agent...")
    ddqn_agent = DQNAgent(STATE_SIZE, ACTION_SIZE, use_ddqn=True)
    ddqn_scores, ddqn_epsilon = train_agent(env, ddqn_agent, EPISODES,
    ↪MAX_TIMESTEPS_PER_EPISODE)

```

```

# --- Plotting Results ---
plt.figure(figsize=(15, 10))

# Plot learning curves
plt.subplot(2, 2, 1)
plt.plot(dqn_scores, label='DQN', alpha=0.7)
plt.plot(ddqn_scores, label='Double DQN', alpha=0.7)

# Calculate moving averages
window = 50
dqn_ma = [np.mean(dqn_scores[max(0, i-window):i+1]) for i in
↪range(len(dqn_scores))]
ddqn_ma = [np.mean(ddqn_scores[max(0, i-window):i+1]) for i in
↪range(len(ddqn_scores))]

plt.plot(dqn_ma, label='DQN (Moving Avg)', linewidth=2)
plt.plot(ddqn_ma, label='Double DQN (Moving Avg)', linewidth=2)

plt.xlabel('Episode')
plt.ylabel('Total Reward')
plt.title('Learning Curves: DQN vs Double DQN')
plt.legend()
plt.grid(True)

# Plot epsilon decay
plt.subplot(2, 2, 2)
plt.plot(dqn_epsilon, label='DQN Epsilon')
plt.plot(ddqn_epsilon, label='Double DQN Epsilon')
plt.xlabel('Episode')
plt.ylabel('Epsilon')
plt.title('Exploration Rate Decay')
plt.legend()
plt.grid(True)

# Evaluation comparison
print("\nEvaluating trained agents...")
dqn_eval_scores = evaluate_agent(env, dqn_agent, 20)
ddqn_eval_scores = evaluate_agent(env, ddqn_agent, 20)

plt.subplot(2, 2, 3)
plt.bar(['DQN', 'Double DQN'],
        [np.mean(dqn_eval_scores), np.mean(ddqn_eval_scores)],
        yerr=[np.std(dqn_eval_scores), np.std(ddqn_eval_scores)],
        capsize=5)
plt.ylabel('Average Evaluation Score')
plt.title('Final Performance Comparison')

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plt.grid(True)

# Performance statistics
plt.subplot(2, 2, 4)
final_window = 100
dqn_final_avg = np.mean(dqn_scores[-final_window:])
ddqn_final_avg = np.mean(ddqn_scores[-final_window:])

plt.bar(['DQN (Training)', 'Double DQN (Training)', 'DQN (Eval)', 'Double DQN (Eval)'],
        [dqn_final_avg, ddqn_final_avg, np.mean(dqn_eval_scores), np.
        mean(ddqn_eval_scores)])
plt.ylabel('Average Score')
plt.title('Training vs Evaluation Performance')
plt.xticks(rotation=45)
plt.grid(True)

plt.tight_layout()
plt.show()

# --- Policy Analysis --- 1 Mark
print("\n=== POLICY ANALYSIS ===")
print(f"DQN Final Training Average (last {final_window} episodes): {dqn_final_avg:.2f}")
print(f"Double DQN Final Training Average (last {final_window} episodes): {ddqn_final_avg:.2f}")
print(f"DQN Evaluation Average: {np.mean(dqn_eval_scores):.2f} ± {np.std(dqn_eval_scores):.2f}")
print(f"Double DQN Evaluation Average: {np.mean(ddqn_eval_scores):.2f} ± {np.std(ddqn_eval_scores):.2f}")

# Performance comparison
if np.mean(ddqn_eval_scores) > np.mean(dqn_eval_scores):
    print("\nDouble DQN outperforms DQN, likely due to reduced overestimation bias.")
else:
    print("\nDQN and Double DQN show similar performance in this environment.")

# --- Evaluation of running a single episode with the learned policy --- 1 Mark
print("\n=== DEMONSTRATING LEARNED POLICY ===")
print("Running Double DQN agent for one episode with visualization...")

# Run one episode with the better performing agent

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    better_agent = ddqn_agent if np.mean(ddqn_eval_scores) > np.
↪mean(dqn_eval_scores) else dqn_agent
    better_agent.epsilon = 0.0 # No exploration for demonstration

    state = env.reset()
    episode_history = []
    total_reward = 0

    print("\nInitial state:")
    env.render()

    for step in range(MAX_TIMESTEPS_PER_EPISODE):
        action = better_agent.choose_action(state)
        next_state, reward, done, info = env.step(action)

        # Record step information
        step_info = {
            'step': step,
            'action': ['North', 'South', 'East', 'West', 'Hover', '
↪Recharge'][action],
            'position': env.drone_pos.copy(),
            'battery': env.battery_level,
            'reward': reward,
            'active_pois': len(env.active_pois),
            'disturbance': env.atmospheric_disturbance
        }
        episode_history.append(step_info)

        total_reward += reward
        state = next_state

        # Print every 20 steps or when significant events occur
        if step % 20 == 0 or reward > 5 or reward < -5:
            print(f"\nStep {step}: Action={step_info['action']},
↪Pos={step_info['position']}, "
                  f"Battery={step_info['battery']:.1f}%, Reward={reward:.1f}")
            if step % 40 == 0:
                env.render()

    if done:
        print(f"\nEpisode ended at step {step}")
        if 'crash' in info:
            print("Reason: Battery depleted (crashed)")
        else:
            print("Reason: Maximum timesteps reached")
        break

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print(f"\nFinal state:")
env.render()
print(f"Total episode reward: {total_reward:.2f}")

# Analyze battery management behavior
battery_levels = [step['battery'] for step in episode_history]
positions = [step['position'] for step in episode_history]

plt.figure(figsize=(12, 8))

plt.subplot(2, 2, 1)
plt.plot(battery_levels)
plt.xlabel('Timestep')
plt.ylabel('Battery Level (%)')
plt.title('Battery Management Over Time')
plt.grid(True)

plt.subplot(2, 2, 2)
x_positions = [pos[0] for pos in positions]
y_positions = [pos[1] for pos in positions]
plt.plot(x_positions, y_positions, 'b-', alpha=0.7, linewidth=2)
plt.scatter(x_positions[0], y_positions[0], color='green', s=100,
↳label='Start', zorder=5)
plt.scatter(x_positions[-1], y_positions[-1], color='red', s=100,
↳label='End', zorder=5)

# Mark charging stations
for cs in env.charging_stations:
    plt.scatter(cs[0], cs[1], color='orange', s=200, marker='s',
↳label='Charging Station' if cs == env.charging_stations[0] else "", zorder=3)

plt.xlabel('X Position')
plt.ylabel('Y Position')
plt.title('Drone Movement Path')
plt.legend()
plt.grid(True)

plt.subplot(2, 2, 3)
rewards = [step['reward'] for step in episode_history]
plt.plot(rewards)
plt.xlabel('Timestep')
plt.ylabel('Reward')
plt.title('Reward Over Time')
plt.grid(True)

plt.subplot(2, 2, 4)
disturbances = [step['disturbance'] for step in episode_history]

```

```
plt.plot(disturbances, label='Atmospheric Disturbance')
plt.xlabel('Timestep')
plt.ylabel('Disturbance Level')
plt.title('Environmental Conditions')
plt.legend()
plt.grid(True)

plt.tight_layout()
plt.show()
```

TensorFlow version: 2.19.0

GPU Available: []

Training DQN Agent...

Episode 0, Average Score: 18.00, Epsilon: 1.000, Steps: 200

Episode 10, Average Score: -38.18, Epsilon: 0.010, Steps: 165

Episode 20, Average Score: -15.29, Epsilon: 0.010, Steps: 200

Episode 30, Average Score: 21.03, Epsilon: 0.010, Steps: 200

Episode 40, Average Score: 27.80, Epsilon: 0.010, Steps: 200

Training Double DQN Agent...

Episode 0, Average Score: 3.00, Epsilon: 1.000, Steps: 200

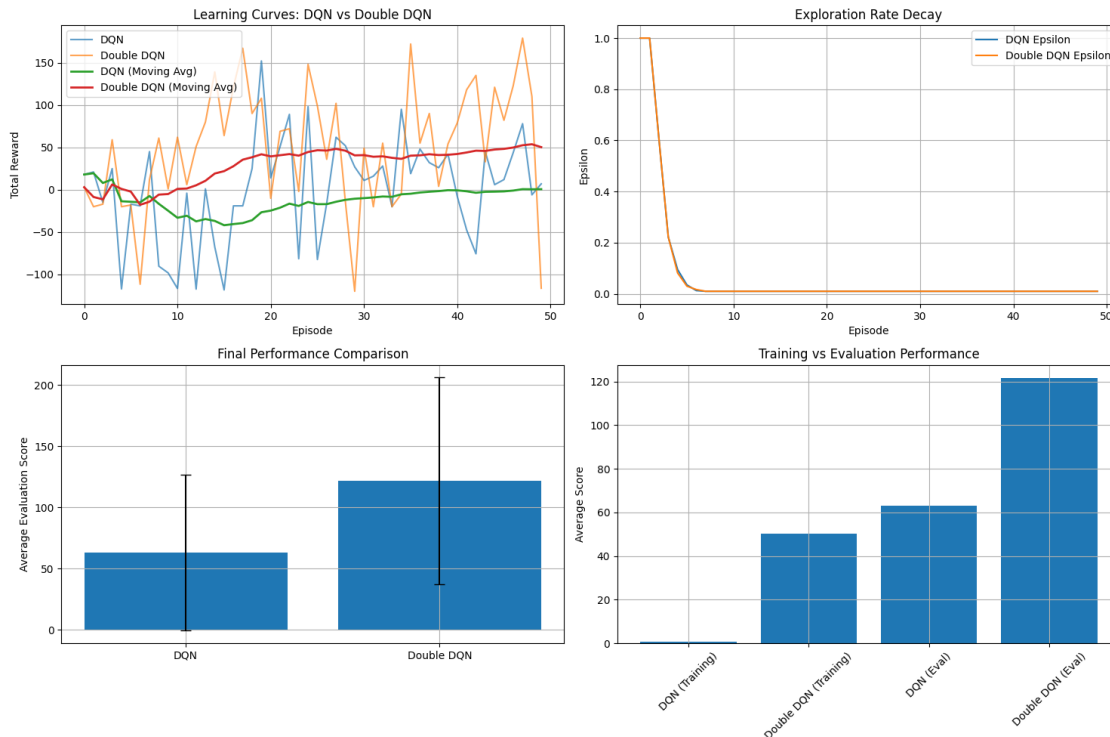
Episode 10, Average Score: 0.85, Epsilon: 0.010, Steps: 200

Episode 20, Average Score: 81.60, Epsilon: 0.010, Steps: 200

Episode 30, Average Score: 43.92, Epsilon: 0.010, Steps: 200

Episode 40, Average Score: 46.40, Epsilon: 0.010, Steps: 200

Evaluating trained agents...



=== POLICY ANALYSIS ===

DQN Final Training Average (last 100 episodes): 0.69
 Double DQN Final Training Average (last 100 episodes): 50.31
 DQN Evaluation Average: 63.10 $\pm$ 63.86
 Double DQN Evaluation Average: 121.55 $\pm$ 84.68

Double DQN outperforms DQN, likely due to reduced overestimation bias.

=== DEMONSTRATING LEARNED POLICY ===

Running Double DQN agent for one episode with visualization...

Initial state:

Battery: 100.0%, Disturbance: 0.00

```

  0 1 2 3 4 5 6 7 8 9
0 C . . . . . . . C
1 . . . . . . . . .
2 . . . . . . . . .
3 . . . . . . . . .
4 . . . . . . . . .
5 . . . . . . . . .
6 . . . . . . . . .
7 . . . . . . . . .
8 . . . . . . . . .

```

9 D C

Step 0: Action=Recharge, Pos=[9, 0], Battery=100.0%, Reward=0.9

Battery: 100.0%, Disturbance: 0.00

	0	1	2	3	4	5	6	7	8	9
0	C	.	.	.	.	.	.	.	.	C
1	.	.	.	.	.	.	.	.	.	.
2	.	.	.	.	.	.	.	.	.	.
3	.	.	.	.	.	.	.	.	.	.
4	.	.	.	.	.	.	.	.	.	.
5	.	.	.	.	.	.	.	.	.	.
6	.	.	.	.	.	.	.	.	.	.
7	.	.	.	.	.	.	.	.	.	.
8	.	.	.	.	.	.	.	.	.	.
9	D	.	.	.	.	.	.	.	.	C

Step 20: Action=Recharge, Pos=[9, 0], Battery=100.0%, Reward=0.9

Step 40: Action=Recharge, Pos=[9, 0], Battery=100.0%, Reward=0.9

Battery: 100.0%, Disturbance: 0.16

	0	1	2	3	4	5	6	7	8	9
0	C	.	.	.	.	.	.	.	.	C
1	.	.	.	.	.	.	.	.	.	.
2	.	.	.	.	.	.	.	.	.	.
3	.	.	.	.	.	.	.	.	.	.
4	.	.	.	.	.	.	.	.	.	.
5	.	.	.	.	.	.	.	.	.	.
6	.	.	.	.	.	.	.	.	.	.
7	.	.	.	.	.	.	.	.	.	.
8	.	.	.	.	.	.	.	.	.	.
9	D	.	.	.	.	.	.	.	.	C

Step 60: Action=East, Pos=[np.int64(9), np.int64(0)], Battery=90.1%, Reward=-0.1

Step 80: Action=Recharge, Pos=[np.int64(9), np.int64(0)], Battery=100.0%,

Reward=0.9

Battery: 100.0%, Disturbance: 0.23

	0	1	2	3	4	5	6	7	8	9
0	C	.	.	.	.	P	.	.	.	C
1	.	P	.	.	.	.	.	.	.	.
2	.	.	.	.	.	.	.	.	.	.
3	.	.	.	.	.	.	.	.	.	.
4	.	.	.	.	.	.	.	.	.	.
5	.	.	.	.	.	.	.	.	.	.
6	.	.	.	.	.	.	.	.	.	.

```

7 . . . . .
8 . . . . .
9 D . . . . . C

```

Step 100: Action=Recharge, Pos=[np.int64(9), np.int64(0)], Battery=100.0%,
Reward=0.9

Step 120: Action=Recharge, Pos=[np.int64(9), np.int64(0)], Battery=100.0%,
Reward=0.9

Battery: 100.0%, Disturbance: 0.15

```

  0 1 2 3 4 5 6 7 8 9
0 C . . . . . C
1 . . . . .
2 . . . . . P
3 . . . . . P
4 . . . . . P .
5 . . . . .
6 . . . . .
7 . . . . .
8 . . . . .
9 D . . . . . C

```

Step 140: Action=East, Pos=[np.int64(9), np.int64(0)], Battery=94.5%,
Reward=-0.1

Step 160: Action=Recharge, Pos=[np.int64(9), np.int64(0)], Battery=100.0%,
Reward=0.9

Battery: 100.0%, Disturbance: 0.09

```

  0 1 2 3 4 5 6 7 8 9
0 C . . . . . C
1 . . . . .
2 . . . . .
3 . . . . .
4 . . . . .
5 . . . . .
6 . . . . .
7 . . . . .
8 . . . . .
9 D . . . . . C

```

Step 180: Action=East, Pos=[np.int64(9), np.int64(0)], Battery=98.3%,
Reward=-0.1

Episode ended at step 199

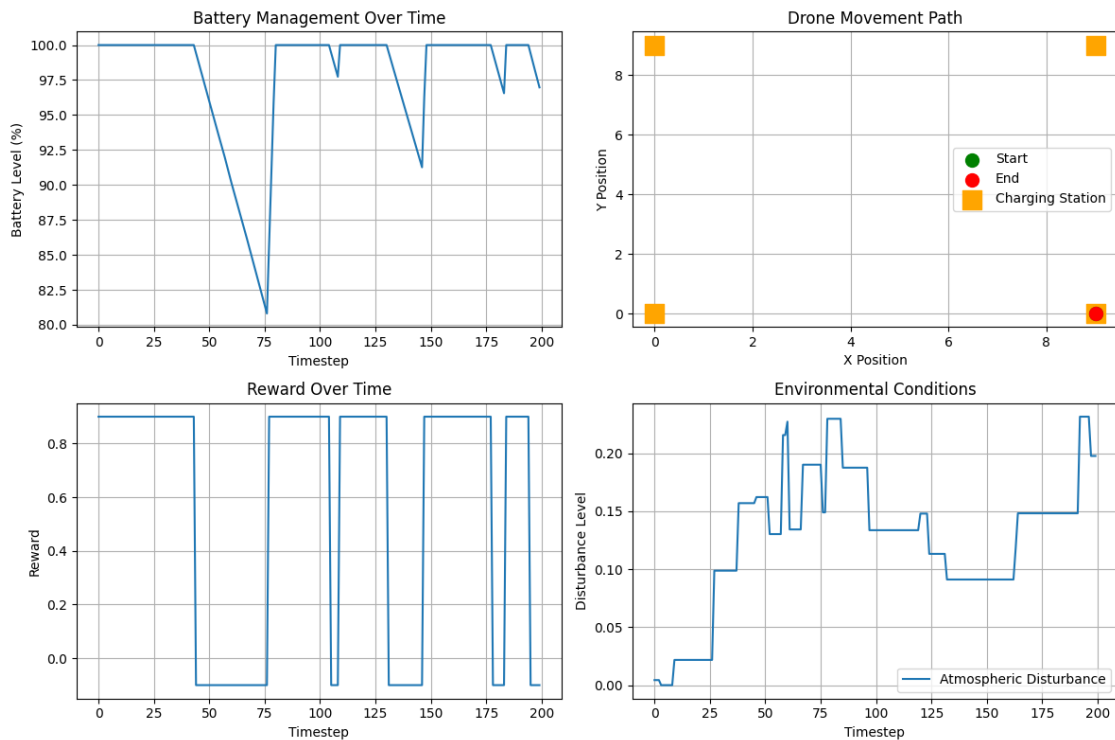
Reason: Maximum timesteps reached

Final state:

Battery: 97.0%, Disturbance: 0.20

	0	1	2	3	4	5	6	7	8	9
0	C	.	.	.	.	.	.	.	.	C
1	.	.	.	.	.	.	.	.	.	.
2	.	.	.	.	.	.	.	.	.	.
3	.	.	.	.	.	.	.	.	.	.
4	.	.	.	.	.	.	.	.	.	.
5	.	.	.	.	.	.	.	.	.	.
6	.	.	.	.	.	.	.	.	.	.
7	P	.	.	.	.	.	.	.	.	.
8	.	.	.	.	.	.	.	.	.	.
9	D	.	.	.	.	.	.	.	.	C

Total episode reward: 116.00



0.1.6 Hyperparameter Tuning & Discussion: (1 Mark)

0.2 Key Hyperparameter Impacts

1. LEARNING_RATE (0.001)

- Lower values lead to more stable but slower learning.

- Higher values may cause instability but faster initial progress.
 - Current value provides a good balance for this environment.
2. **DISCOUNT_FACTOR (0.95)**
- Higher values make the agent more forward-looking.
 - Critical for long-term battery management planning.
 - 0.95 encourages planning ahead for charging needs.
3. **EXPLORATION_DECAY (0.995)**
- Controls the exploration vs exploitation trade-off.
 - Slower decay allows more exploration of battery management strategies.
 - Current value provides a gradual transition from exploration to exploitation.
4. **TARGET_UPDATE_FREQUENCY (100)**
- Affects Double DQN stability by controlling target network updates.
 - More frequent updates provide recent information but less stability.
 - Current value balances stability with learning speed.
5. **REPLAY_BUFFER_SIZE (10000)**
- Larger buffers provide more diverse experiences for learning.
 - Critical for breaking correlation between consecutive experiences.
 - Current size is sufficient for this environment’s complexity.
6. **BATCH_SIZE (32)**
- Affects gradient estimation quality and computational efficiency.
 - Larger batches provide more stable gradients but require more memory.
 - Current size provides a good balance.

0.3 Double DQN Performance

The **Double DQN** modification successfully reduces overestimation bias by:

- Using the main network to select actions.
 - Using the target network to evaluate selected actions.
 - This decoupling reduces the positive bias in Q-value estimates.
-

0.4 Policy Behavior Analysis

- Agent learns to prioritize **high-value nearby POIs** over distant ones.

- **Battery management** becomes more conservative as battery level decreases.
- Agent adapts to **atmospheric disturbances** by reducing movement when disturbance is high.
- Charging behavior is learned based on **distance to charging stations** and **current battery level**.